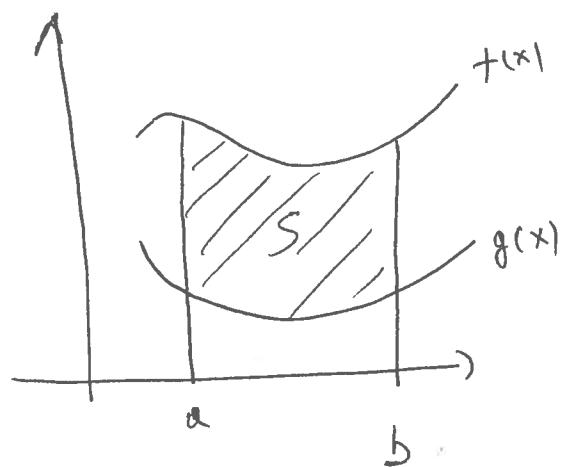
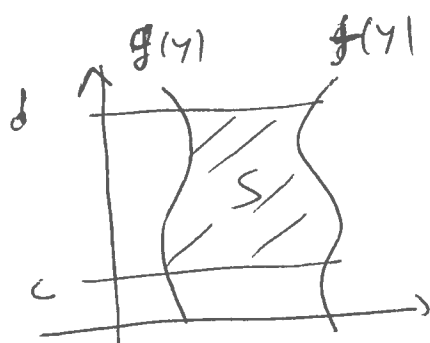


13.3 Double integrals on non-rectangular regions



y simple set

$$S = \{ (x, y) \mid g(x) \leq y \leq f(x), a \leq x \leq b \}$$



x -simple set

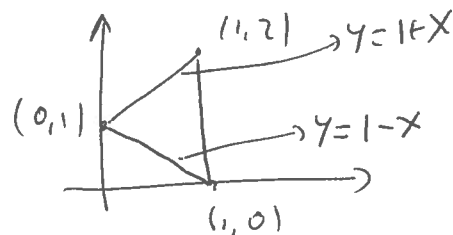
$$S = \{ (x, y) \mid g(y) \leq x \leq f(y), c \leq y \leq d \}$$

Suppose S is y -simple as above, then

$$\iint_S F(x, y) dA = \int_a^b \left[\int_{g(x)}^{f(x)} F(x, y) dy \right] dx$$

Eg S = triangle with vertices $(0, 1)$, $(1, 0)$, $(1, 2)$

$$\begin{aligned} \iint_S x+y dA &= \int_0^1 \int_{1-x}^{1+x} (x+y) dy dx \\ &= \int_0^1 \left[xy + \frac{y^2}{2} \right]_{1-x}^{1+x} dx = \int_0^1 2x^2 + 2x dx \\ &= \left[\frac{2}{3}x^3 + x^2 \right]_0^1 = \frac{2}{3} + 1 = \frac{5}{3} \end{aligned}$$



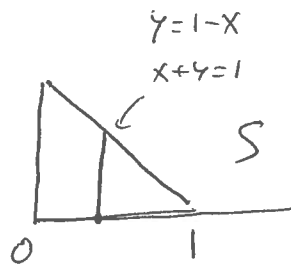
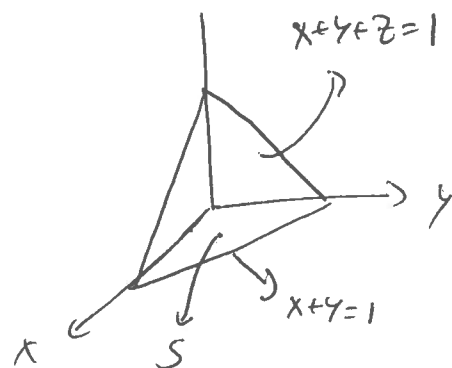


$$\iint_S y \cos x \, dA$$

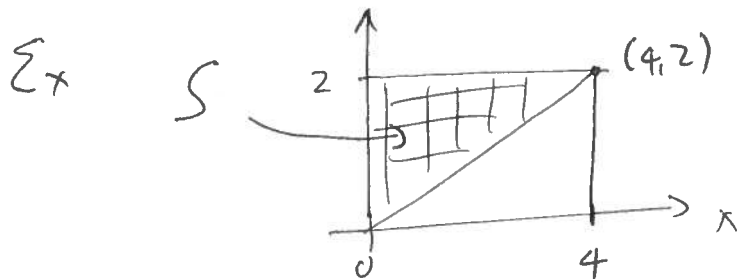
$$\begin{aligned} \int_0^1 \left[\int_0^{y^2} y \cos x \, dx \right] dy &= \int_0^1 \left[y \sin x \right]_0^{y^2} dy = \int_0^1 y \sin y^2 \, dy = \\ &= \left[-\frac{\cos y^2}{2} \right]_0^1 = -\frac{(\cos 1)}{2} + \frac{(\cos 0)}{2} = \frac{1 - \cos 1}{2} \end{aligned}$$

Ex Compute the volume of the pyramid with vertices $(0,0,0)$ $(0,1,0)$ $(1,0,0)$, $(0,0,1)$

$$\begin{aligned} \iint_S 1-x+y \, dA &= \int_0^1 \int_0^{1-x} (1-x-y) \, dy \, dx \\ &= \int_0^1 \left[y - xy - \frac{y^2}{2} \right]_0^{1-x} dx = \int_0^1 (1-x)^2 - \frac{(1-x)^2}{2} \, dx \\ &= \int_0^1 \frac{(1-x)^2}{2} \, dx = \left[-\frac{(1-x)^3}{6} \right]_0^1 = \frac{1}{6} \end{aligned}$$



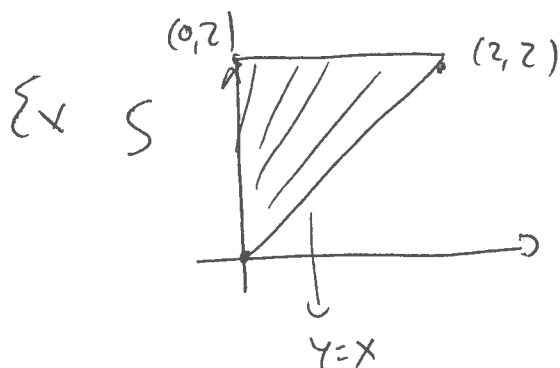
Note Volume = $\frac{\text{Area base} \times \text{height}}{3} = \frac{\frac{1}{2} \times 1}{3} = \frac{1}{6}$



$$\iint_S e^{y^2} dA$$

$$\int_0^4 \int_{x/2}^2 e^{y^2} dy dx \quad ?? \text{ too hard!}$$

$$\begin{aligned} \int_0^2 \int_0^{2y} e^{y^2} dx dy &= \int_0^2 [x e^{y^2}]_0^{2y} dy = \int_0^2 2y e^{y^2} dy \\ &= [e^{y^2}]_0^2 = e^4 - 1 \end{aligned}$$



$$\iint_S \frac{z}{1+x^2} dA \rightarrow \int_0^2 \int_0^y \frac{z}{1+x^2} dx dy$$

$$\int_0^2 \int_x^2 \frac{z}{1+x^2} dy dx$$

$$\int_0^2 2 \tan^{-1} x \Big|_0^y dy$$

$$\int_0^2 2 \tan^{-1} y dy$$

|| ← HARD

$$\left[2y \tan^{-1} y - \frac{z}{2} \ln(1+y^2) \right]_0^2$$

$$\begin{aligned} & \left[4 \tan^{-1} x \right]_0^2 = \left[\ln(1+x^2) \right]_0^2 \leftarrow \int_0^2 2 \frac{2-x}{1+x^2} dx \\ & 4 \tan^{-1} 2 - \ln 5 \end{aligned}$$

$$4 \tan^{-1} 2 - \ln 5$$

Ex 1 compute a) $\int_1^3 \int_{-y}^{2y} x e^{y^3} dx dy$

b) $\int_1^5 \int_0^x \frac{3}{x^2+y^2} dy dx$ c) $\int_0^{\pi/4} \int_{\sqrt{2}}^{\sqrt{2} \cos \theta} r dr d\theta$ d) $\int_0^2 \int_0^{\sqrt{4-x^2}} (x+y) dy dx$

e) $\iint_S xy dA$ S = region bounded by $y=x^2$ & $y=1$

f) $\iint_S (x^2+2y) dA$ S = region between $y=x^2$ & $y=\sqrt{x}$

g) $\iint_S \frac{z}{1+x^2} dA$ S = triangle with vertices $(0,0)$ $(2,2)$ $(0,2)$

h) $\iint_S x dA$ S bounded by $y=x$ $y=x^3$